Recursive Scientific Governance as a Falsifiable Decision Contract

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Abstract

Research agents can generate hypotheses, retrieve literature, design experiments and draft interpretations faster than they can reliably determine what those outputs scientifically warrant. We present **ORION-RSE**, a fail-closed recursive scientific-governance contract that separates research generation from claim promotion. ORION-RSE atomizes claims, subtracts nearest-donor ownership before novelty promotion, freezes protected discriminators before outcomes, retains null/negative/subsumed history, distinguishes standalone from interaction-only evidence, records `CANNOT_CHECK`, separates evaluator identity from scientific authority, stops recursion when no material discriminator remains and reopens prior decisions only after material evidence or regime change. P14A is retained as a negative mixed benchmark because the decisive discriminator occupied only 1.8375% of realized cases and aggregate gates could not pass; we then show that negative is uninformative rather than adverse, since both failing gates read one quantity whose supremum over the protocol's own declared sampling support is 0.042326, below both the 0.05 and the 0.08 bar, so its seven-gate conjunction had a single reachable value before the seed was drawn. The terminal is retained verbatim and reclassified `CANNOT_CHECK`. P14B creates a balanced semantic discriminator but is diagnostic because the full policy reused the same adjudication function as gold and because hostile review found a protocol conformance mismatch. P14C is the specification-separated successor: 28 explicit gold cases are frozen separately, gold/rationale/case identity are stripped before policy calls, the full implementation is independent, and six component ablations are tested. Full ORION-RSE is 28/28 correct with zero false promotions and full valid-discovery recall; `MULTI_REVIEW` is 24/28 with 14.29% false promotion, all ablations are worse, and two evaluations produce identical canonical SHA-256 `74 032348de7e6508b6c1827aabcf1bf9d354d30b9c6f81c8259fdb3535f01a63`. P14C's terminal is reachable in both directions — of the seven implementations its protocol admits in the graded slot, exactly one clears every gate — and P14A's 0.05 and 0.08 bars, registered unchanged, are both met there at 0.142857. The evidence supports a precise controlled claim: **against the registered governance specification, the complete contract conforms strictly better than partial review contracts without suppressing valid promotion.** External scientific validity remains open.

1 Introduction and donor boundary

Scientific research is not equivalent to producing a plausible hypothesis or a positive result. A claim must be interpreted relative to prior work, the question frozen before observation, negative/null evidence, competing explanations, resource constraints and evaluator authority. Research agents amplify both opportunity and failure: they can produce more candidate studies, but also more false novelty, widened claims, post-hoc endpoints, forgotten negatives, donor duplication and confident closure when evidence is non-identifying.

Autonomous and semi-autonomous systems already search literature, generate hypotheses, run code, design experiments and revise research artifacts. Goal-evolving scientific agents make recursive discovery an explicit objective. Reflection, self-critique, reviewer agents and multi-agent debate own iterative review; preregistration/checklists own prospective endpoint discipline; truth-maintenance systems own dependency-aware history; provenance owns evidence lineage; authorization systems own separation of actor permission from operation.

ORION-RSE claims none of those primitives. Its question is:

> **Can an explicit scientific-governance contract make better claim-promotion decisions than strong partial review contracts under matched information and decision resources, while preserving valid discoveries?**

The live residual is the **scientific promotion contract plus its evaluation**: claim atomization, donor subtraction, protected freeze, negative/subsumed history, interaction/non-identifiability dispositions, authority separation, recursion stops and material-change reopening composed into one fail-closed lifecycle and evaluated on false-promotion, disposition and useful-discovery endpoints.

Within the ORION programme, the disposition semantics this contract consumes are frozen upstream donors (per `papers/SYNC_CONTRACT.md`): ORION-16 owns transition-admissibility / disposition semantics, ORION-17 owns regime-change reopen semantics, ORION-18 owns authorization typing and discharge, and ORION-15 owns the evaluator-custody/authority-separation rule instantiated here. ORION-RSE's claim is the decision-process result — that the composed fail-closed governance process makes better promotion decisions at matched capability — not novelty in the authorization semantics themselves. Execution-integrity receipts of the ORION-25 class are admissible as evidence substrate for packet provenance/replay, with their dispositions imported as frozen inputs rather than re-adjudicated.

2 ORION-RSE contract, lifecycle and evaluation

A research atom carries a bounded claim/question, parent identity, nearest donors, protected discriminator, protocol identity, evidence/resource receipts, evaluator identity, authority owner, disposition and stop/reopen conditions. Atomization is an investigation unit, not automatic publication fragmentation.

The contract distinguishes `SUPPORTED_RESIDUAL`, `SUBSUMED`, `INTERACTION_ONLY`, `REDUNDANT_EQUIVALENT`, `OVERREACH_HARMFUL`, `NON_IDENTIFIABLE`, `CANNOT_CHECK`, negative/null outcomes and `RETAIN_NEGATIVE` when new material does not justify reopening. Positive observation is therefore not identical to positive scientific promotion.

Before novelty promotion, nearest prior mechanisms are recorded and the residual is classified as adopted, adapted, composed, deferred or subsumed. The discriminator and terminal rules are frozen before protected outcomes; a material post-outcome change requires a new protocol identity. Negative results can motivate successors but cannot be rewritten.

The lifecycle is:

`question -> atoms -> donor subtraction -> protected discriminator -> protocol freeze -> execution/receipts -> disposition -> recurse/stop -> later reopen if materially warranted`.

`CANNOT_CHECK` is a valid scientific disposition when evidence or authority is insufficient.

2.1 Evaluation contract

All policies receive identical case facts and identical fixed decision-check budgets. Registered policies are `RAW_POSITIVE`, `REFLECTION_CHECKLIST`, `DONOR_AWARE_REVIEW`, `MULTI_REVIEW`, `ORION_RSE_FU`

LL and component ablations. Primary safety endpoint is false scientific promotion; productivity endpoint is useful-discovery recall among gold SUPPORTED_RESIDUAL cases; full disposition accuracy and component-specific correctness are secondary. This prevents an always-abstain system from winning.

3 P14A — preserved negative mixed benchmark

P14A used 20 held-out families × 400 cases with independently varied fact rates. Full ORION-RSE produced false promotion 0.000000, useful-discovery recall 1.000000, disposition accuracy 1.000000 and history/reopen accuracy 1.000000.

The strongest comparator, MULTI_REVIEW, produced false promotion 0.018375, useful-discovery recall 1.000000, disposition accuracy 0.981625 and history/reopen accuracy 0.505051.

The frozen protocol required strongest-baseline false promotion ≥0.05 and accuracy separation ≥0.08. Both gates failed. Terminal: P14A_CONTROLLED_GOVERNANCE_SUPERIORITY_GATE_NOT_MET. This terminal is permanent.

3.1 Root cause

After validity, freeze, identifiability, donor and interaction checks, the only difference between MULTI_REVIEW and full ORION-RSE is live negative history without material new evidence. In the realized random mixture, that effective discriminator occupied only **1.8375%** of cases, so the maximum possible aggregate accuracy gap was also 1.8375 points. The result identifies a benchmark-design problem: natural/random mixtures may underweight the decision boundary being tested. P14A is not retuned; it motivates a balanced successor.

3.2 What the negative established, and what it did not

The paragraph above is a statement about the draw that happened. The stronger statement is about every draw the protocol admits, and it is the one that decides how the terminal may be cited.

Both failing gates read one quantity. MULTI_REVIEW reproduces the gold adjudication except where a positive is a same-evidence rereading of a live negative history; over the 144 fact states the generator can emit, that exception is a single state, and it is also MULTI_REVIEW's only false promotion. strongest_baseline_false_promotion_ge_0_05 reads its frequency; accuracy_gain_ge_0_08 reads 1 - (1 - that frequency). They are one number, which is why the receipt prints 0.018375 twice.

The eight case facts are independent Bernoulli draws whose rates are each monotone in a different declared uniform, and every family's draw is mixed half-and-half with a fixed base. The prevalence is therefore a product whose extrema over the declared box sit at corners, and that box's **supremum is 0.042326** — below the 0.05 bar and well below the 0.08 one. Exercised over five registered admissible worlds spanning the declared ranges and ending at the extremal corner, the best reachable value is 0.040250, so the attainment margins are **−0.009750** and **−0.039750** and no admissible world satisfies either gate. The other five gates are satisfied by every admissible world — three of them because the graded ORION_RSE_FULL arm is the gold function that scores it, measured at 0 divergent points of 256. The seven-gate conjunction therefore had **one reachable value** before the seed was drawn.

The emitter is not the problem. Re-opening the declared sampling ranges and nothing else — same seed, same nine arms, same seven thresholds, same terminal expression — moves the terminal to the positive branch in three of three registered capability worlds, with two distinct terminals observed and no inert case.

P14A's terminal, seed, thresholds and receipt are retained verbatim and nothing is relabelled positive. Its evidential disposition is `CANNOT_CHECK`: a measurement the frozen protocol could not take, not evidence against the governance contract. The adjudication is `P14_GATE_ATTAINABILITY_ADJUDICATION_V1.json`, and the successor benchmark answers the same question at these unchanged thresholds.

A threshold is a claim about a distribution. Freeze the threshold and the support of the statistic it reads together, or an outcome has been frozen rather than a test.

4 P14B — balanced semantic discriminator, diagnostic only

P14B uses a fresh seed and 12 held-out families with seven equal strata: clean support, legitimate material reopening, retained negative history, donor subsumption, interaction-only evidence, `CANNOT_CHECK` and negative evidence. There are 6,720 protected cases total with the same seven-check receipt for every arm.

Full ORION-RSE yields zero false promotion, 1.0000 disposition accuracy and 1.0000 useful-discovery recall. `MULTI_REVIEW` yields 0.142857 false promotion, 0.857143 disposition accuracy and 1.0000 useful-discovery recall. Both systems correctly allow genuinely material reopening; only full ORION-RSE retains negative history when no material new evidence exists. Every registered component ablation is worse.

P14B is not authoritative primary evidence. First, its full policy originally invoked the same decision function used as protected gold, making the implementation comparison circular. Second, hostile PR review found the realized generator did not implement the protocol's promised nuisance reminting across all strata. The benchmark is retained as semantic-development history and carries no primary claim authority.

5 P14C — specification-separated primary benchmark

P14C was frozen after the P14B circularity issue was identified and does not alter P14A or P14B.

- `P14C_ADJUDICATION_CASES_V1.json` contains **28 explicit frozen cases**, four variants for each of seven semantic strata.
- `run_p14c_specification_separated_governance_v1.py` implements every policy independently from the case table.
- Before a policy call, the harness strips `gold_disposition`, `rationale`, `case_id` and `stratum`; the policy receives only factual booleans.
- Precedence variants test donor-over-interaction/history, interaction-over-history, validity failure and negative evidence.
- Six ablations separately remove evidence-integrity, freeze, identifiability, donor, interaction and negative-history checks.

The explicit specification is internal to the programme, not a human external-adjudication dataset. P14C therefore evaluates **conformance to a separately frozen specification**, not truth about open-ended science.

Terminal: `P14C_SPECIFICATION_SEPARATED_GOVERNANCE_CONFORMANCE_SUPPORTED`.

policy	accuracy	false promotion	discovery recall
ORION*RSE*FULL	**1.0000**	**0.0000**	**1.0000**
MULTI_REVIEW	0.857143	0.142857	1.0000
DONOR_AWARE_REVIEW	0.714286	0.285714	1.0000
REFLECTION_CHECKLIST	0.571429	0.428571	1.0000
RAW_POSITIVE	0.428571	0.535714	1.0000

Full ORION-RSE correctly handles all retained-negative and supported-reopen cases. All six ablations lower disposition accuracy. Gold is absent from every policy input. Two independent evaluations produce identical canonical SHA-256 `74032348de7e6508b6c1827aabcf1bf9d354d30b9c6f81c8259fdb3535f01a63`.

5.1 Could this terminal have been the other word?

The question that disqualified P14A's negative is asked of P14C's positive, over the coordinate P14C leaves free. P14A sampled the discriminator's prevalence; P14C fixes it — four cases per semantic stratum makes the `RETAIN_NEGATIVE` share exactly `4/28 = 0.142857` in every admissible table, against P14A's ceiling of `0.042326` for the same quantity. What stays free is the implementation in the graded slot, and the protocol registers seven: the full contract and its six single-component ablations.

Over that register exactly one subject clears all eight gates and six fail at least one, so the conjunction prints **two** distinct terminals. `accuracy_advantage_ge_0_10` fails for the donor, interaction and negative-history ablations, so it is refutable rather than assumed. `strongest_baseline_false_promotion_ge_0_10` is a benchmark-difficulty precondition and holds in every admissible table by construction — the property P14A's corresponding certificate lacked, where it was unsatisfiable instead.

P14A's two frozen bars, registered verbatim and read here, are both met:

- `strongest_baseline_false_promotion_ge_0_05`: reachable interval [0.142857, 0.142857] over admissible tables; realized 0.142857.
- `accuracy_gain_ge_0_08`: reachable interval [0.000000, 0.142857] over admissible subjects; realized 0.142857.

The 0.08 bar sits strictly inside its interval, so it could have gone either way. The question P14A's gates encoded is answered affirmatively at P14A's own thresholds, under P14C's protocol identity, with nothing of P14A edited.

One residual is reported rather than absorbed: `full_discovery_recall_one` is satisfied by all seven registered subjects, because an ablation removes a check and a policy reading fewer facts promotes more rather than fewer. No registered implementation abstains, so that gate carries no refutation capacity here and its pass is not evidence that the contract preserves valid discovery.

The strongest current claim is: **against an explicit adjudication specification frozen separately from policy implementation, the full ORION-RSE contract conforms to every registered governance case and strictly outperforms the registered partial-governance implementations without reducing valid promotion.** This is stronger than a self-call conformance test and weaker than external scientific validity.

6 Required external validation and resource accounting

A broad claim that ORION-RSE improves real research requires prospectively assembled realistic packets across multiple domains such as ML experiments, formal/software verification and computational science. Candidate workflows should receive identical literature/source packets, tools and model budgets. Independent blinded adjudicators must determine the maximum admissible claim, donor overlap, negative-evidence handling, `CANNOT_CHECK` and valid discoveries.

Comparators should include a strong research agent, reflection/self-critique, researcher–reviewer multi-agent workflow, literature-RAG plus experiment planning, preregistration/checklist control and ORION component ablations.

Longitudinal evaluation should create positive, null, negative and subsumed history in round 1, then change evidence, donor set or regime in round 2. The system must avoid both forgetting old negatives and over-transferring old negatives after genuine material change. Useful-discovery noninferiority remains co-primary so safety cannot be purchased by blanket abstention.

Every workflow should report model/checkpoint, evidence access, web/literature access, tool access, generated tokens, experiment/search budget, reviewer/evaluator calls, number of research/reflection passes and end-to-end latency. ORION-RSE cannot hide extra compute as "governance." Final scientific authority remains external to candidate workflows.

7 Relation to current research-agent work, limitations and conclusion

Goal-evolving and autonomous scientific systems establish open-ended generation as donor-owned. Research on scientific-exploration behavior shows that evaluation should consider what agents choose to explore, not only whether they can produce plausible papers. ORION-RSE studies a complementary layer: **scientific admissibility governance**. Its distinctive mechanism is active negative/subsumed history plus material-reopen semantics on top of validity, donor and interaction checks.

P14A–C are controlled symbolic decision benchmarks, not natural-language open-ended science. P14C removes direct gold-function reuse but its specification is still internally authored; external validity remains open. Balanced strata test discrimination, not real-world prevalence; P14A shows why prevalence and discriminability must be reported separately. Real workflows have heterogeneous costs. Human or independent-model adjudicators may disagree on novelty and admissibility. Longitudinal value of negative-history retention is not yet shown on realistic research. Poorly calibrated reopen/materiality criteria could suppress productive risk-taking, so useful-discovery noninferiority is mandatory. No claim of frontier autonomous-research superiority is authorized.

ORION-RSE treats scientific governance as a decision system that can itself fail. P14A fails because its decisive discriminator is too rare; P14B is semantically informative but non-authoritative; P14C removes implementation circularity with a separately frozen case specification and independent policy implementation. The complete contract conforms to all registered cases, makes zero false promotions and preserves every valid promotion, while the strongest partial review contract fails the retained-negative boundary.

The next claim frontier is external rather than synthetic: **blinded realistic packets, independent adjudication and longitudinal research history under matched resources.**

8 References

- Du, Y. et al. *Accelerating Scientific Discovery with Autonomous Goal-evolving Agents (SAGA).* arXiv:2512.21782.
- Tang, Y. & Yang, Y. *AI Research Agents Narrow Scientific Exploration.* arXiv:2605.27905, 2026.
- Related reflection, multi-agent review/debate, preregistration, truth-maintenance, provenance and authorization literatures are donor-owned foundations and are discussed in the integrated manuscript.